2025 Formula SAE-A Technical Inspection Sheet
ACCUMULATOR INSPECTION

Initial Presentation Scrutineer Name:		**Initial Presentation Lane Number:**	
1st Reinspection Scrutineer Name:		**1st Reinspection Lane Number:**	
2nd Reinspection Scrutineer Name:		**2nd Reinspection Lane Number:**	
3rd Reinspection Scrutineer Name:		**3rd Reinspection Lane Number:**	

PERSONNEL Teams are required to have an Electrical Safety Officer.			**Initial ins.**	**Inspection #** **1st reins.**	**2nd reins.**	**3rd reins.**
AI.1.1	Identify the Electrical Safety Officer (ESO)	Ask for the ESO				
AI.1.2	Only team members involved in technical inspection can be in the technical inspection bay	Ask for non-essential people to leave the area				

DO NOT PROCEED WITH TECHNICAL INSPECTION IF THE ESO IS NOT PRESENT

INSTRUCTIONS FOR SCRUTINEERS

This section applies to Electric Vehicles (EV) and Autonomous Vehicles (AV) and Dual Purpose Vehicles.
If an item is acceptable, initial the "Passed" column.
If an item is unacceptable, initial the "Reinspect" column and inform the team members they have not meet the requirements for that item.
Teams are allowed to take their vehicle away from technical inspection and address any issues before representing for reinspection.
Scrutineers may continue with the inspection when unacceptable items are found, provided it is safe to do so.
Teams are required to pass all items before the vehicle can proceed to the Brake Test or compete in Dynamic Events.

Scrutineers reserve the right to refuse to proceed with Technical Inspection if a sufficiently serious issue which might jeopardize the safety of the technical inspection team, event personal or any other persons is found at any time.

- ESO (Lukes), Oscar, Ryan, Jayson and Malcolm (charger), Bunny (document hander), Cam (SES)
- Charger, Accumulator (race ready) bolted to the accumulator trolley,
- Accumulator spares box
- Red toolbox, callipers and large cable crimps (blue handle)
- Tech Box which samples, documents and tech bibles
- 2 x HV Mats
- 4 x Safety Glasses, 3 x LV Gloves and 3 x Leather outers
- HV Insulated tools – Socket sets, Spanners, Screwdrivers, cutter, shears, Allen keys
- Fluke Multimeter with probes and banana jack proves

EV CLASS - ACCUMULATOR INSPECTION

SAFETY EQUIPMENT AND TOOLS

Teams are expected to have appropriate safety equipment and tools to allow them to work on their vehicle.
All safety equipment and tools must be in good usable condition and comply to Australian industry best practice standards.
There must be a sufficient quantity of appropriate safety equipment (PPE) for everyone who needs to work on electrical systems.
Insulated tools must be rated and appropriately marked as complaint to DIN/IEC 60900 (VDE 1000V), or an equivalent recognized standard.
Improvised or repaired tools using heat shrink, tape or similar are not acceptable. Modified tools are not acceptable.

				Reinspect	Passed
AI.2.1	Safety Glasses	At least one pair per team member working on electrical systems. Safety Glasses should be non-conductive (non-metallic). Display safety glasses which should be laid out, know where safety rating is	Inspect		
AI.2.2	HV Insulating Gloves	HV Isolating gloves with leather outers. Gloves must comply with AS2225 or ASTM D120 or AS IEC 60903 The inner gloves must have been tested in the last 12 months and be free from damage or punctures. Outer gloves must be in serviceable condition. Gloves must be labelled with a working voltage, higher than the vehicle's maximum battery voltage. At least one pair per team member working on electrical systems. Size of the gloves should be appropriate for those conducting the work. Note: Class 00 gloves are rated for up to 500V, Class 0 gloves are rated for up to 1000V. Lay gloves out separated from the outers	Inspect 	p.g.1	
AI.2.3	HV Isolating Blanket	Two HV Isolating Blanket (at least 0.83 square meter) Measure or look at data sheet Blanket must be in good condition and free from punctures or tears. Blanket must be labelled with a working voltage, higher than the vehicle's maximum battery voltage.	Inspect	p.g.1-2	
AI.2.4	HV Insulated Tools	Check the team has an appropriate set of tools to maintain the vehicles electrical systems. These must include as a minimum all of the following: - Insulated cable shear Display - Insulated screwdrivers Display - Insulated pliers Display - Multimeter with protected probe tips (CATIII 600V or better) Display - A pair of quality insulated 5mm banana test leads (600V rated or higher) Display	Inspect	p.g.3	
AI.2.5	Additional Insulated Tools	If the accumulator uses bolted connections the team must also present tools appropriate for manipulating the selected fasteners such as: - Insulated spanners Display - Insulated socket set Display - Insulated Allen keys (hex wrenches) Display - Any other special tools required to maintain the accumulator and other electrical systems safely	Inspect	p.g.3	

EV CLASS - ACCUMULATOR INSPECTION

CHARGING TROLLEY				Reinspect	Passed
AI.3.1	TSACHC Wheels	The hand cart must have at least two wheels. Appropriately rate to take the load of the accumulator and cart. Display datasheet, the accumulator is 70kg, trolley weight is much less than 70 kg	Inspect	p.g.4	
AI.3.2	TSACHC Brake	The hand cart must have a brake which acts on at least 2 of the wheels that is always on and only released if someone pushes the handle, or similar. The brake must be capable of safely stopping the fully loaded hand cart. Display by moving the trolley and operating the brake lever	Inspect		
AI.3.3	TSACHC Accumulator Mounting	Accumulator must be securely attached to the hand cart to enable safe transportation. Display trolley with accumulator bolted to the base.	Inspect		
AI.3.4	TSACHC Labels	The label on the TSACHC must include The vehicle number, the university name, and the ESO phone number(s) must be displayed and written in Roman characters of at least 10mm high on the lid or top of each TSACHC. The characters must be clearly visible and placed on a high-contrast background. Display labels on Trolley, Use a ruler to show height of letters	Inspect		

BATTERY CHARGER INSPECTION Only chargers presented and sealed at technical inspection are allowed.				Reinspect	Passed
AI.4.1	Battery Charger	Charger needs to be professionally built. All connections of the charger(s) must be insulated and covered. No open connections are allowed. Surloks must be covered, Malcolm to be present and to explain that he built it and is a qualified electrician	Inspect		
AI.4.2	Battery Charger Wiring	Battery Charger Power lead must be free of any damage, nicks, cuts, grease or grime. The plug shall be suitably rated for the current load and conforming to EN 60309-2, AS/NZS 3123, Charger shall have a valid test and tag sticker, or tag attached. Display Datasheet, display testing tag and conforming sticker	Inspect	p.g.5	
AI.4.3		HV wires must be marked with gauge, temperature rating and voltage rating, serial number or norm is also sufficient, if the team shows the datasheet in printed form. Display samples and datasheets	Inspect	p.g.6-10	
AI.4.4		Wire temperature and voltage rating must be suitable for use in the charger Display Datasheet	Inspect	p.g.7-10	
AI.4.5		Using only insulating tape or rubber-like paint for insulation is prohibited. We do not use	Inspect		
AI.4.6		The AC power supply to the battery charger and other associated devices must include a residual current device (RCD) with over current protection (fuses or an appropriate circuit breaker) or residual current circuit breaker (RCBO). The RCD or RCBO device must act to disconnect both the active and neutral supplies. The trip sensitivity of the RCD must not exceed 30mA. Where possible 10mA is preferred. Display Datasheet and show on charger. Display schematic	Inspect	p.g.11-13	
AI.4.7		Traction System charging leads must be orange. Display HV Charging Cable	Inspect	p.g.8	

EV CLASS - ACCUMULATOR INSPECTION

BATTERY CHARGER INSPECTION CONTINUED... Only chargers presented and sealed at technical inspection are allowed.				Reinspect	Passed
AI.4.8	Charging system measuring points	Two charging system voltage measuring points must be provided on the charger output lines. Display Schematic and Test Points	Inspect	p.g.13	
AI.4.9		The measuring points must be protected from being touched with the bare hand/fingers once the housing is opened. Display Test points with covers	Inspect		
AI.4.10		4mm shrouded banana jacks rated to a voltage higher than the maximum tractive system voltage must be used Display datasheet	Ask the team to provide a datasheet	p.g.14	
AI.4.11		The Charger System Measurement Points must be protected by body protection resistors per EV6.8.4 Display schematic, datasheet and be prepared to demonstrate using FLUKE by doing the HV+ test point and outlet and the HV- test point and outlet. They should both be 15kΩ	Inspect	p.g.15	
AI.4.12		The Charger System Measuring Points must be marked with HV+ and HV- Display test points	Inspect		
AI.4.13	Battery Charger Emergency Stop	The charger must include a push-type emergency stop button which has a minimum diameter of 25mm and must be clearly labelled. The Pushbutton must be easy to reach with the accumulator in place. Display Estop and the length of the accumulator charging cables. Use callipers to measure button size	Inspect		
AI.4.14		When the E-stop is pressed, the output from the charger must be electrically disconnected from the battery, and the output from the charger must fall to less than 60 V in 5 seconds. Display video	Inspect		
AI.4.15		The international electrical symbol consisting of a red spark on a white-edged blue triangle must be affixed in close proximity to this switch Display sticker	Inspect		
AI.4.16	Battery Charger Earthing	The battery charger, accumulator and accumulator charging trolley, and any associated metallic components must be equipotential bonded and connected to the AC power supply earth such that the RCD function is not impeded. Cables used for earthing must be yellow/green stripped and at least 2.5mm^2 cross section. Display photos and earth cable connecting everything, display datasheet	Inspect	p.g.16-17	
AI.4.17	Battery Charger Galvanic Isolation	The Charger DC output must be galvanically isolated from the AC input display datasheets	Inspect	p.g.18	

EV CLASS - ACCUMULATOR INSPECTION

ACCUMULATOR INSPECTION Accumulator technical inspection will inspect the accumulator outside of the vehicle in the designated battery room. Teams are expected to bring the necessary tools for the safe opening of the accumulator for inspection. All accumulators will be physically inspected.				Reinspect	Passed
AI.5.1	Accumulator Construction	HV Accumulator(s) must be enclosed in container(s). Display	Inspect		
AI.5.2		The bottom of the accumulator must be steel of minimum thickness 1.25mm or Aluminium of minimum thickness 3.2mm. Walls (internal and external) and cover/top, must be Steel of minimum thickness 0.9mm or aluminium of minimum thickness of 2.3mm. If alternative material used, they should present proof of equivalency (per SES or other) with a sample. Display photos and/or measure the physical container	Inspect	p.g.19	
AI.5.3		All components and parts of the accumulator container need to be properly fixed. Internal cables and components must not be able to move around causing chafing, stress to connections, or other damage. Display, wiggle and touch cables	Inspect		
AI.5.4		Fasteners in the tractive system high current path must have a positive locking mechanisim. The locking mechanism must be suitable for use at a minimum operating temperature of 90 degrees C. Bolts with nylon patches are acceptable only for blind holes and where used per OEM design. Any bolt which may pose a risk of fire or short circuit if loosened must utilise a positive locking mechanism. Display that only genlocks are used in the accumulator which are rated to 300 degrees C	Inspect		
AI.5.5		The accumulator cells must be secured from moving in the event of an impact. This includes if the accumulator in inverted. Generally, friction fitting is not sufficient, and some positive retention is required. If the accumulator lid is used as a method to retain the cells, it must be reinforced and robustly attached. Display calc of lid strength and that the top plate is reinforced. Display image of cells under top plate	Inspect	p.g.20 - 22	
AI.5.6		Internal vertical walls must be robustly fastened to the container Display welds on the outside of the container and the bottom, show photos	Inspect	p.g.19	
AI.5.7		If the accumulator container is made from an electrically conductive material, the poles of the accumulator stack(s) and/or cells must be insulated against the inner wall of the accumulator container, if the container is made of electrically conductive material. There must be an insulating barrier between the accumulator walls and any electrically live component within. This usually means that all of the internal walls of the accumulator need to be lined with an appropriate insulating material. Display photos and lining itself and the datasheet for the FR4, the CANaMONS are made from FR4 and protect it from above, as well as the GPO-3 on the Top Plate	Inspect	p.g.23	
AI.5.8		Holes in the accumulator external structure should be of the minimum practical size for cable passages, plugs etc. Holes for ventilation should be circular, no greater than 10mm in diameter, and not obviously weaken the accumulator container external structure. When installed in the vehicle, the holes must not face towards the driver with the firewall removed. Display photos and measure using Vernier Callipers	Inspect	p.g.19	

EV CLASS - ACCUMULATOR INSPECTION

ACCUMULATOR INSPECTION CONTINUED...				Reinspect	Passed
AI.5.9	Labelling	Accumulator must be marked with the ISO7010-W012 symbol triangle with black lightning bolt on yellow background) with triangle side length of 100 mm minimum, with text "ALWAYS ENERGIZED" If accumulator voltage is greater than 60VDC must also be Labelled as "HIGH VOLTAGE" Labelling should be on all practical external sides including top, bottom, front, rear, sides. Smaller labels accepted only if surface too small for defined size. Display Labelling on accumulator, measure using ruler	Inspect		
AI.5.10	Internals - cell connection	Contacting / interconnecting the single cells by soldering in the high current path is prohibited. Soldering wires to cells for the voltage monitoring input of the BMS is allowed Display photo, we do not use solder	Inspect	p.g.20	
AI.5.11		Connections to cell tabs must be adequately supported to prevent damage to the cells. Cables, busbars etc. must not be supported entirely by the cell tabs. Display photo, bolted to push bar	Inspect	p.g.24	
AI.5.12	Internals – IR/fuse	Every accumulator container must contain at least one main fuse and at least two accumulator insulation relays. There must be no electrically live connection to the accumulator cells outside of the accumulator enclosure when the IRs are open. This includes BMS wires. IRs must be rated for the maximum expected current draw. The IRs and main fuses must be separated from the cell segments by an electrically insulated and non-flammable material. Display the datasheets for the fuse and IRs and point out the High current Path. Display that they are separated by the top plate and the GPO-3 datasheet	Inspect	p.g.25-27	
AI.5.13	Internals – fuses	The fuse(s) must be appropriately rated to protect the cable(s) and other electronic components. Fuses should be HRC type and marked with a DC voltage rating above the maximum expected operating voltage. Except where specifically stated in the rules, all small wiring connecting directly to the cells or tractive system should have appropriate fuses as close as possible to the source. Fusible links are acceptable but require evidence of appropriate voltage and current rating. Teams may justify undersized (current only) fuses based on average current draw and the manufacturer's thermal time curves. Display datasheet for fuses, and CANaMONS schematic to show fuses are used for the voltage taps, all fuses use 22 AWG or larger which is rated for 7 A, our fuses are 0.5 A. Datasheet for CANaMONS fuses is under the schematic.	Inspect	p.g.28-29	
AI.5.14	Internals - maintenance plugs	Maintenance plugs or similar measures have to be taken to allow separating the internal cell stacks in a way, that the separated cell stacks carry a voltage of less than 120VDC and a maximum energy of 12MJ. The separation has to affect both poles of the stack. It must not be possible to incorrectly install the maintenance plugs causing a short circuit. Display the service handle and how it is bidirectional and separates the pack into 9 segments. 1 Segment = 2.80MJ and 50.4 V maximum. Also display the tap covers we use. Display Methode Bud Connector Datasheet	Inspect	p.g.30	
AI.5.15	Internals – cell stack barriers	Each stack has to be electrically insulated by the use of suitable material towards other stacks in the container and on top of the stack. Air is not considered to be a suitable insulator for this purpose. The team should provide evidence that the material(s) used are appropriate (i.e. material datasheets, manufacturer's specifications, lab test reports etc.). Team manufactured composite materials are not acceptable (carbon fibre, Kevlar, fiberglass etc.). Display datasheets for FR4 and GPO3	Inspect	p.g.23-27	

EV CLASS - ACCUMULATOR INSPECTION

ACCUMULATOR INSPECTION CONTINUED...				Reinspect	Passed
AI.5.16	Accumulator container connectors	If HV-connectors of the accumulator containers can be removed without the use of tools, a pilot contact/interlock line has to be implemented which breaks the current through the IRs whenever the connector is removed Display connector cover and how to requires a tool to remove	Inspect		
AI.5.17	Indicator light/voltmeter	Each container must have an indicator light or an analogue voltmeter showing that voltages greater than 60V DC are present outside of the container. This includes when the accumulator is being charged. Display on lid and how it is connected to IRs	Inspect		
AI.5.18	Equalizing valve	If the container is completely sealed, it must have an equalizing valve to prevent integral build-up of pressure in the event of fire. N/A	Inspect		
AI.5.19	Electrical Connections	Electrical connections must be secure and of high quality. Connections must be made using appropriate hardware, correct sized lugs for selected cable, appropriate crimping methods etc. Electrical connections must not be able to loosen if they become hot. Electrical connections which rely on plastic or similar material being squeezed, which might soften and become loose if overheated are not acceptable. Information about the electrical connections supporting the high current path must be available at Elect Tech Inspection. Display Datasheets for Methode Buds, Radloks, and Surloks. Display Blue handle crimps and an example of a crimped lug	Inspect	p.g.30-32	
AI.5.20	Precharge and Discharge Resistors	Precharge and discharge resistors must be installed such that if they overheat, it is unlikely they will cause any of the accumulator cells to overheat. If the team has elected to utilize Precharge or discharge or discharge resistors with a continuous operation (not monitored by the PDOC per local addendum), the resistors must install according to the manufacturer's recommendations for heatsinking airflow, and such that heat soak cannot cause the accumulator cells to overheat. We do not have continuously operating Precharge or discharge resistors. Show datasheets for the Precharge and discharge resistor, only the Precharge resistor is in the accumulator. Discharge is in the HIP. The Precharge is only on when the accumulator is recharging, which is no more than a couple second and it will not overheat (controlled by teensy activating a Precharge relay)	Inspect	p.g.33-34	
AI.5.21	Accumulator Internal Wiring	All wires inside the accumulator must be marked with gauge, temperature rating and voltage rating, serial number or norm is also sufficient, if the team shows the datasheet in printed form. Display datasheets and samples	Inspect	p.g.35-36, pg.8	
AI.5.22		Wire temperature and voltage rating must be suitable for use in the accumulator. All Wires (including GLV and BMS) wires within the accumulator container must be rated for the full tractive system voltage. Display datasheets and samples	Inspect	p.g.35-36	
AI.5.23		Using only insulating tape or rubber-like paint for insulation is prohibited. N/A	Inspect		

EV CLASS - ACCUMULATOR INSPECTION

ACCUMULATOR INSPECTION CONTINUED...				Reinspect	Passed
AI.6.1	Plastic and 3D printed Components	Any plastic components which are critical to the integrity of the accumulator (structural elements, bolt locking devices, insulation, barriers etc.) must be manufactured from materials which have a glass transition temperature higher than the expected operation temperature. The operating temperature is considered to be the calculated maximum cable or busbar temperature, maximum battery temperature or pre-charge or discharge resistor operating temperature, or 90 degrees C whichever is higher. 90 degrees C is the highest temperature as our cables and bus bars are over specced, maximum battery temperature is 60 degrees C and the Precharge resistor is only in operation during Precharge. Display FireWire datasheet and operating temperatures for bus bars and cable	Inspect	p.g.37-38	
AI.6.2	Spare accumulator(s)	Spare Accumulators must be of the same type (construction, voltage, capacity, weight, fixing etc.) Only applicable if spare accumulators are used. N/A	Inspect		
AI.6.3	Temperature monitoring equipment	Teams will be provided with a thermally sensitive sticker to install into their accumulator. The thermally sensitive sticker must be mounted such that it is direct thermal contact with the negative most cell of an accumulator segment such that it the thermal sticker will reasonably represent the temperature of the cells. The sticker should be visible from outside the accumulator without needing disassembly of the accumulator. (2025 only - refer inaccessible stickers to Technical Inspection Captain for review) During the fitment of the thermal strip, take photos to show and ensure that it is visible through the hatch in the accumulator	Inspect		

EV CLASS - ACCUMULATOR INSPECTION

NOTES

INSTRUCTIONS FOR SCRUTINEERS - END OF EV ACCUMULATOR INSPECTION

If the team have passed all items in this section, go to the second page of this document and initial the relevant box, noting the date and time the team passed this section.

If the team have items requiring reinspection (did not pass all items), go to the second page of this document and initial the relevant box noting the time and date the team was released from the inspection. The team will need to rectify the reinspection items and may return for reinspection when they are ready. Reinspection is completed on a first come, first served basis, and is dependent on Technical Inspection lane availability. Teams with Scheduled Inspections times will be given priority over teams presenting for reinspection.

Once the team have been released from the inspection, make note of the team's progress on the central inspection record board.

2025 Formula SAE-A Technical Inspection Sheet
EV STATIC INSPECTION

Initial Presentation Scrutineer Name:		**Initial Presentation Lane Number:**	
1st Reinspection Scrutineer Name:		**1st Reinspection Lane Number:**	
2nd Reinspection Scrutineer Name:		**2nd Reinspection Lane Number:**	
3rd Reinspection Scrutineer Name:		**3rd Reinspection Lane Number:**	

PERSONNEL Teams are required to have an Electrical Safety Officer.			**Initial ins.**	**Inspection #** **1st reins.**	**2nd reins.**	**3rd reins.**
SI.1.1	Identify the Electrical Safety Officer (ESO)	Ask for the ESO				
SI.1.2	Only team members involved in technical inspection can be in the technical inspection bay	Ask for non-essential people to leave the area				
DO NOT PROCEED WITH TECHNICAL INSPECTION IF THE ESO IS NOT PRESENT						

INERT VEHICLE The Vehicle must be in a condition which prevents unexpected energization of the HV electrical systems or movement of the tractive system			**Initial ins.**	**Inspection #** **1st reins.**	**2nd reins.**	**3rd reins.**
SI.1.3	Disable the HV Systems - Manual Service Disconnect (MSD)	Check the MSD is removed				
SI.1.4	Disable the HV Systems - Tractive System Master Switch (TSMS) Lock	Check the TSMS Lock is fitted and effective				
SI.1.5	(AV and Dual Purpose only) Disable the ASMS - Autonomous System Master Switch (ASMS) Lock	Check the ASMS Lock is fitted and effective				
DO NOT PROCEED WITH TECHNICAL INSPECTION IF MSD IS INSTALLED OR THE TSMS LOCK IS MISSING						

INSTRUCTIONS FOR SCRUTINEERS

This section applies to Electric Vehicles (EV) and Autonomous Vehicles (AV) and Dual Purpose Vehicles.
If an item is acceptable, initial the "Passed" column.
If an item is unacceptable, initial the "Reinspect" column and inform the team members they have not meet the requirements for that item.
Teams are allowed to take their vehicle away from technical inspection and address any issues before representing for reinspection.
Scrutineers may continue with the inspection when unacceptable items are found, provided it is safe to do so.
Teams are required to pass all items before the vehicle can proceed to the Brake Test or compete in Dynamic Events.

Scrutineers reserve the right to refuse to proceed with Technical Inspection if a sufficiently serious issue which might jeopardize the safety of the technical inspection team, event personal or any other persons is found at any time.

EV CLASS - EV STATIC INSPECTION

- ESO (Lukes), Jayson, Bunny, Mia (document hander) and Cam (SES)
- MSD Disconnected (In ESO’s possession)
- TSMS Lock Installed (Keys in ESO’s possession)
- Car in Race Ready Condition (accumulator installed, everything connected, firewall and body kit on
- Push Bar
- High Stands x 2
- Red SCA Jack Stands x 4
- Red toolbox, callipers, spare zipties, DT crimps and large cable crimps (blue handle)
- Tech Box which samples, documents and tech bibles
- Fluke Multimeter with probes and banana jack proves
- Multiple hands to lift the car and remove the body kit
- Fisher ready with SES open on LAPTOP

Roll car into bay and immediately ask if the car can be put on the stands, leave body kit on

After MSD display take body kit off and place onto stands

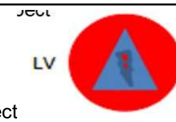
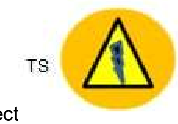
EV CLASS - EV STATIC INSPECTION

VEHICLE VISUAL INSPECTION

The primary purpose of the Vehicle Visual Inspection is to find and highlight any items which might cause a safety hazard for the drivers, team members, track marshals and anyone else who might interact with the vehicle.
The hazards most associated with EV tractive systems include fire, arc flash, electric shock or loss of control. The Visual Inspection is intended to check that the vehicle has been designed and constructed to an acceptable standard such that the likelihood of one of these hazards causing injury is minimized. Scrutineers are not generally concerned with the reliability of the vehicle unless the failure of a component or system might pose a safety hazard. Scrutineers may also look for rules compliance issues. Rules non-compliances should be referred to the EV Technical Inspection Captain. Because of the bespoke nature of the vehicles, Scrutineers reserve the right to reject vehicles for any reasonable safety concern based on their judgement, even if not specifically covered in this document.

				Reinspect	Passed
SI.2.1	Race Ready Condition	The vehicle should be fully assembled with the Accumulator in place and generally in race ready condition when presented to technical inspection. Inspection of incomplete vehicles may proceed with permission from the Technical Inspection Captain. Ensure Accumulator is in, and everything is connected (without HV switch and MSD). Body kit and top fire wall must be fitted	Inspect		
SI.2.2	Technical Inspection Access	The vehicle must be lifted/jacked onto stands such that all of the vehicle's wheels are at least 100 mm and no more than 300mm above the ground. It must be possible to freely rotate the vehicle's wheels without the vehicle moving. Stands must be robust, and the vehicle must be secure. Use red SCA car jack stands on the first height setting to support the car in the air. Use a tape measure to display its height off the ground from the wheels. Body Kit Stays on	Inspect		
SI.3.1	Push Bar	A pair of HV insulating gloves with leather outers, a Multimeter and fire extinguisher must be attached. If a tool is needed to open the MSD this must also be attached to the push bar. Unlock box on the push bar to display the HV gloves with leather outers, a Multimeter and a fire extinguisher. The HV inners should be tagged from Accumulator Tech and the Fire Extinguisher must be in date	Inspect		
SI.4.1	Manual Service Disconnect	The MSD must be clearly marked with "MSD" in large, high contrast lettering and should be clearly visible to a Marshal as they approach the vehicle. The MSD label must be easily distinguishable from the vehicle's other markings. Display MSD labels on Body Kit and HIP	Inspect		
SI.4.2		It must be possible to disconnect the MSD without removing any bodywork Display MSD location	Inspect		
SI.4.3		If opening the MSD is possible without the use of tools, a pilot contact/interlock line has to be implemented which breaks the current through the IRs whenever the connector is removed Display schematic for the Shutdown Circuit and HIP, Display physical MSD	Inspect	p.g.1	
SI.4.4		In ready to race condition it must be possible to disconnect the MSD within 10 seconds. Ask the scrutineer if it is okay to fit the MSD to demonstrate. Fit the MSD and remove it once the inspector is ready. After this is done take body kit off	Ask the team to demonstrate		
SI.5.1	Tractive system protection	All tractive system connections outside of an enclosure must be appropriately insulated. Display HV Cables from the accumulator, the HIP and the MC, Display the Phase Cable Cover	Inspect		
SI.5.2		It must not be possible to touch any tractive system connections with a 100mm long, 6mm diameter insulated test probe when the tractive system enclosures are in place Ensure test points covers are on. Let the inspector probe the car, comply to their requests	Inspect		
SI.5.3		Tractive System components and containers must be protected from moisture in the form of rain or puddles. Use of tape alone is not acceptable. Heat shrink must be appropriately rated for the TS voltage and must be provided with additional mechanical protection. Display Heat Shrink datasheet and samples and say that is it only used on the phase cable cover and in the accumulator	Inspect	p.g.2	

EV CLASS - EV STATIC INSPECTION

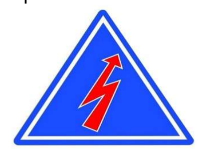
VEHICLE VISUAL INSPECTION CONTINUED…				Reinspect	Passed
SI.6.1	Master Switches	The Tractive System Master Switch (TSMS), Grounded Low Voltage Master Switch (GLVMS) and (AV and Dual Purpose only) Autonomous System Master Switch (ASMS) on the right side of the vehicle. At the height of the drivers' shoulders. Rotary type, ON position with key horizontal. ON and OFF position must be clearly marked. Located right side of the vehicle at approx. driver's shoulder height, just behind main roll hoop. Not attached to removable bodywork. Must be easily actuated from outside the vehicle. Display the MS location and all the above details	Inspect		
SI.6.2		All master switches must be a rotary type with a red removable key/handle Display keys and switches	Inspect		
SI.6.3		GLVMS (LV Master switch) must be completely surrounded by an RED circle of min 50mm diameter Marked with international symbol (triangle with red lightning bolt on blue background). Display	 Inspect		
SI.6.4		TSMS (TS Master switch) must be completely surrounded by an ORANGE circle of min 50mm diameter Accompanied by the ISO 7010-W012 (triangle with black lightning bolt on yellow background). Display	 Inspect		
SI.6.5		The Tractive System Master Switch must have a lock or similar fitted which prevents insertion of the master key. The lock should be of the type commonly used in industry for "LOTO isolation". Use of a LOTO hasp or similar is acceptable, if necessary, provided it is secure. The key for the lock and the master keys should be in the custody of the ESO. Display lock on TSMS and the keys in the possession of the ESO	Inspect		
SI.7.1	HV warning stickers	Each housing/enclosure containing HV parts must be marked with the ISO7010-W012 symbol triangle with black lightning bolt on yellow background) If accumulator voltage is greater than 60VDC must also be Labelled as "HIGH VOLTAGE" Labelling should be on all practical external sides including top, bottom, front, rear, sides. Smaller labels accepted only if surface too small for defined size. Display Stickers on HIP, DCDC, Accumulator, Phase cable cover and Inverter (inverter will be hard to see)	Inspect Check top, sides and bottom of enclosures		

EV CLASS - EV STATIC INSPECTION

VEHICLE VISUAL INSPECTION CONTINUED…				Reinspect	Passed
SI.8.1	HV Wiring	All tractive system wiring outside electrical enclosures must be enclosed in separate orange nonconductive conduit or use an orange shielded cable Display that the 25 mm2 ACC to HIP and the 16 mm2 HIP to DCDC are enclosed in conduit and the HIP to MC and MC to Motor cables are 50 mm2 shielded cable. Show Datasheets	Inspect	p.g.3-5	
SI.8.2		The conduit or shielded cable must be securely anchored (at least) at each end so that it can withstand a force of 200 N without straining the cable and crimp and must be located out of the way of possible snagging or damage. Take note of connections to motors, plugs carrying HV, and any entrances to accumulators, motor controllers, the TSMPs and any other HV enclosures Display boots connecting the conduct to the connector on the ACC to HIP and HIP to DCDC, and that the HV wires are routing in a way that does not show possible snagging or damage.	Inspect		
SI.8.3		Tractive system wiring must be shielded against damage by rotating and/or moving parts. All wiring must be sufficiently restrained such that it cannot come into contact with rotating and/or moving parts. Display that the HV wires are routing in a way that does not show possible snagging or damage.	Inspect		
SI.8.4		TS wires and GLVS wires are clearly separated/do not run directly next to each other/bounded together by cable rods or in the same cable channel (ALLOWED ONLY FOR PILOT CONTACTS OR INTERLOCK SIGNALS) Display that no HV wires run together with LV wires	Inspect		
SI.8.5		Using only insulating tape or rubber-like paint for insulation is prohibited. N/A	Inspect		
SI.8.6		Heat-shrink type insulation used on tractive system components and cables may only be used inside rigid enclosures and must be labelled with voltage rating. Serial number or norm is also sufficient, if the team shows the datasheet in printed form. Display heat shrink datasheet and samples,	Inspect	p.g.2	
SI.8.7		No soldering in high current path including cable lugs and connectors. We do not use solder in the high current path, everything is exclusively bolted, crimped or connected using a connector.	Inspect		
SI.8.8		Electrical connections must be secure and of high quality. Connections must be made using appropriate hardware, correct sized lugs for selected cable, appropriate crimping methods etc. Electrical connections must not be able to loosen if they become hot. Electrical connections which rely on plastic or similar material being squeezed, which might soften and become loose if overheated are not acceptable. Information about the electrical connections supporting the high current path must be available at Elect Tech Inspection. Display Datasheets for HPK, Deutsch, and Surloks. Display Blue handle crimps and an example of a crimped lug, display Deutsch Crimps, HPK come precrimped. DCDC HV connector is under the Surloks	Inspect	p.g.6-8	
SI.8.9		HV wiring must be entirely within the chassis (side impact structure, rear impact zone and roll envelope) except for cables to hub motors Display Phase cable guide and that the rest of the HV wires are within the chassis	Inspect		

EV CLASS - EV STATIC INSPECTION					
VEHICLE VISUAL INSPECTION CONTINUED…				Reinspect	Passed
SI.9.1	Tractive system measuring points	Two tractive system voltage measuring points and a GLVS ground point must be installed next to the master switches, right side of the vehicle, at shoulder height of the driver. Display Test Points	Inspect		
SI.9.2		The measuring points must be protected by a non-conductive waterproof housing which can be opened without tools. The use of simple plastic/rubber sealing plugs inserted into the measuring points is not acceptable as a waterproofing method. Display banana jack covers	Inspect		
SI.9.3		The measuring points must be protected from being touched with the bare hand/fingers once the housing is opened. Take of banana jack covers and display	Inspect		
SI.9.4		4mm shrouded banana jacks rated to a voltage higher than the maximum tractive system voltage must be used Display Datasheet	Ask the team to provide a datasheet	p.g.9	
SI.9.5		The TSMPs must be marked with HV+ and HV- Display markings next to test points	Inspect		
SI.10.1	GLV Ground Measuring Point	Must be positioned next to the TSMPs and must be marked with GND Display GLVMP and GND label	Inspect		
SI.10.2	GLV Voltage	Measure GLVS voltage between GLVS battery positive Must be less than 60V DC Measure using Fluke, measuring between GND and 12V	Ask team to demonstrate		
SI.10.3	Tractive System Voltage	Measure TS voltage at measurement points Must be less than 60V DC Measure using Fluke, using TSMPS	Ask team to demonstrate		
SI.10.4	Discharge System	The discharge circuit has to be wired in a way that it is always active whenever the shutdown circuit is open. If a discharge circuit is used a low resistance can be measured between HV+ and HV- whenever the tractive system is de-activated. Measure resistance between HV+ and HV- with multi-meter. Must be 2*BPR+ Dis- Charge Resistor (GLVS must be off) Use the Fluke to measure the resistance between HV+ and HV- and show the schematic for the HIP and TSAL Discharge, explain that SHUTDOWN_TSMS activates the coil of the discharge relay (datasheet attached) and disconnects the discharge resistor (datasheet also attached) when HV is on. This number must be 2*15KΩ + 5kΩ = 35kΩ Refer alternative discharge methods to the EV Technical Inspection Captain	Ask team to demonstrate	p.g.1	
SI.11.1	Insulation Measurement Test	The insulation resistance between the GLV Ground Measuring Point and TSMPs HV+ must be greater than 500 Ohms/Volt, calculated for the maximum TS voltage. The available measurement voltages are 250 V and 500 V. All vehicles with a maximum nominal operation voltage below 500 V will be measured with the next available voltage level. All teams with a system voltage of 500 V or more will be measured with 500 V. Use the Fluke to measure the resistance between HV+ and GND. This number must be more than 500Ω/V*453.6V = 226.8kΩ	Measure insulation resistance between GLV Ground and HV+		
SI.11.1		The insulation resistance between the GLV Ground Measuring Point and TSMPs HV- must be greater than 500 Ohms/Volt, calculated for the maximum TS voltage. The available measurement voltages are 250 V and 500 V. All vehicles with a maximum nominal operation voltage below 500 V will be measured with the next available voltage level. All teams with a system voltage of 500 V or more will be measured with 500 V. Use the Fluke to measure the resistance between HV- and GND. This number must be more than 500Ω/V*453.6V = 226.8kΩ	Measure insulation resistance between GLV Ground and HV+		

EV CLASS - EV STATIC INSPECTION

VEHICLE VISUAL INSPECTION CONTINUED…				Reinspect	Passed
SI.12.1	Shutdown Buttons	One shutdown button, push-pull or push-rotate-pull on each side behind the driver's compartment (height approx. driver's head), one in the cockpit and easily accessible by the driver in any steering wheel position while wearing gloves and wrist restraints Display photo and positions of estops	Inspect	p.g.10	
SI.12.2		Minimum diameter of shutdown buttons on the side is 40mm. Minimum diameter of shutdown button in the cockpit is 24mm Measure using callipers	Inspect		
SI.12.3		The shutdown buttons are not allowed to be easily removable, e.g. mounted onto a removable body work Display	Inspect		
SI.12.4		All shutdown buttons marked with international symbol Display	Inspect 		
SI.12.5	Cockpit Shutdown Button	Cockpit master switch Pull-On, Push-OFF, or Twist-Pull-On, Push-OFF alongside & unobstructed by steering wheel, easily reached by driver. Minimum 24mm diameter. Marked with international symbol Display	Inspect	p.g.10	
SI.12.6	Brake Over Travel Switch	Brake over travel switch must be secured and positioned behind the brake pedal such that it will be tripped before the master cylinders bottom out or any other stops are reached. The brake over travel switch must be robustly mounted and not be bent or damaged if an over-travel occurs. Display BOTS and then photo and video of it operating	Inspect If necessary, ask team to demonstrate by bleeding the brake circuit	p.g.11	
SI.12.7	Inertia switch	The device must be mechanically attached to the vehicle. It must be possible to demount the device so that its functionality can be tested by shaking it. Mounts including double sided tape, Velcro or similar are not acceptable. Display inertia switch, demount using snips to cut the zip tie and shake to test its functionality, Remount with a zip tie.	Inspect		

EV CLASS - EV STATIC INSPECTION					
VEHICLE VISUAL INSPECTION CONTINUED…				**Reinspect**	**Passed**
SI.13.1	Accelerator Pedal Position Sensor (APPS)	Accelerator Pedal must promptly return to original position, if not actuated Display by pressing accelerator pedal with hand	Inspect		
SI.13.2		At least two sensors must be fitted as APPS and must have separate supply, ground and signal lines from the plausibility detection device Display the connectors going to the pedal and how they lines are separate, display the schematic of the PEN	Inspect	p.g.12	
SI.13.3		The accelerator pedal must have a positive stop to prevent sensors from being mechanically overstressed Display Positive stop on pedal	Inspect		
SI.13.4		Two springs must be used to return the throttle pedal to the off position and each spring must work with the other disconnected Display spare accelerator pedal, show the separate springs	Inspect		
SI.13.5	Brake System Encoder	A brake pedal position sensor or brake pressure switch must be fitted to check for plausibility Display brake pressure sensors and the schematic of the PEN	Inspect	p.g.12	
SI.13.6	Brake Master Cylinders	The brake system master cylinder must be actuated directly or by a mechanical connection. Use of Bowden cables or push-pull Bowden cables is not allowed. The first 90% of the brake pedal travel may be used to regenerate brake energy without actuating the hydraulic brake system. The remaining brake pedal travel must directly actuate the hydraulic brake system, but brake energy regeneration may remain active. Display that the master cylinders are directly actuated for 100% of the brake pedal travel. We do use regen but only between 0 and 10% of the accelerator pedal	Inspect		
SI.14.1	Firewalls (EV specific requirements)	A firewall must separate the driver compartment from components of tractive system (including HV wiring). Brake parts and chassis as conductive paths are allowed to protrude through firewall. See T.1.9.3 Outboard wheel motors and associated cables and other components of the suspension members are excluded from this requirement. Display that only the rear brake line and LV wires pass through the firewall, reference rule: T.1.8.4 b. 'Grommets must be used to seal any pass through for wiring, cables, etc'	Inspect		
SI.14.2		The side of the firewall facing the tractive system must be aluminium with a minimum thickness of 0.5mm and must be deliberately earthed to the chassis ground point. Display CAD image and then display ground strap from the DCDC to the Firewall	Inspect	p.g.13	
SI.14.4		The firewall's integrity must not be compromised by forces from the driver. I.e. the driver must not sit directly against the firewall and the driver's seat must not rely on the structural integrity of the firewall unless it is adequately reinforced. Display images from CAD and FEA	Inspect	p.g.13	

EV CLASS - EV STATIC INSPECTION

VEHICLE VISUAL INSPECTION CONTINUED...				Reinspect	Passed
SI.15.1	Team built PCBs	Where tractive system and GLV are on the same circuit board, teams must be prepared to demonstrate appropriate spacing. Separation space on the circuit boards must be clearly marked. The minimum allowable clearances are: Voltage / Over Surface / Thru Air (cut in board) / Under Conformal Coating 0-50 V DC / 1.6 mm / 1.6 mm / 1 mm 50-150 V DC / 6.4 mm / 3.2 mm / 2 mm 150-300 V DC / 9.5 mm / 6.4 mm / 3 mm 300-600 V DC / 12.7 mm / 9.5 mm / 4 mm If necessary, teams may provide manufacturer's datasheets for components demonstrating appropriate voltage ratings where the components require smaller than allowed clearances. Display the datasheet for the Conformal Coating, samples of the PCBs (without components) and the Conformal coating. Have callipers ready. Show LV/HV Separation Book	Inspect	p.g.14	
SI.16.1	Accumulator Construction and Attachment	If containers are monocoque, they must be mounted with Steel backing plates of at least 2mm thickness. N/A	Inspect		
SI.16.2		The Accumulator Container must also have a minimum number of attachment points, dependent on weight and use minimum Grade 8.8 metric 8mm bolt at each 20 kg = 4 attachments. 20-30 kg = 6 attachments. 30-40 kg = 8 attachments. >40 kg = 10 attachments. Our Accumulator is greater than 40kg (69kg), sample bolt and photo of accumulator attachment points.	Inspect	p.g.15	
SI.16.3	Accumulator Crush Zone Clearance	Ensure 25mm clearance from Side Impact Structure. Ask to see SES. Display CAD image and SES	Inspect	p.g.16	
SI.16.4		If rear mounted: At least 25mm clearance from Rear Impact Structure or any non-crushable items.N/A	Inspect		
SI.16.5		Ensure 25mm clearance of surface from the firewall. Display CAD image	Inspect	p.g.16	
SI.16.6		Non-crushable items behind Rear Impact Structure cannot pass through the structure. Display that the radiator and pumps are too large to pass through the rear impact structure	Inspect		
SI.16.7		Rear Impact protection extends to or pass the upper height of the SIS and is supported to the SIS. Display CAD image	Inspect	p.g.16	

EV CLASS - EV STATIC INSPECTION

ENERGY METER				Reinspect	Passed
SI.17.1	Energy Meter	Energy Meter fitted and secured to the chassis in a location that does not interfere with the mechanical or electrical operation of the vehicle, in accordance with the instructions. Energy Meter must be mounted in protected and readily accessible position, mounted securely and identifiable, so that the operation light can be visually inspected, and the unit can be quickly removed in Parc Ferme. Display energy meter placement in reference to the instruction document, explain how we will remove it and that it will be easily inspected	Inspect		
SI.17.2		Confirm retaining ring on all plugs on energy meter is fully engaged (ring must be clicked over detent). Display	Inspect		
SI.17.3		Check the battery voltage cable is installed with the correct polarity (positive is white, negative is black) and the current sensor is orientated correctly (arrow shows direction of conventional current flow i.e. away from battery if installed on positive lead). Display that we have placed the energy meter is on the positive line and meets the requirements	Ask team to demonstrate		

EV CLASS - EV STATIC INSPECTION

VEHICLE GROUNDING

It is expected that all conductive components within 100mm of the Tractive system will be appropriately grounded (electrically bonded to the GLV system ground).

1. By equipotential bonding the conductive parts within the vehicle, the hazard posed by touching a voltage gradient is reduced, and
2. By ensuring a conductive path back to the chassis ground common point, the likelihood of the IMD correctly detecting the fault and shutting off the vehicle is increased, reducing the period of time for which the hazard can be present.

Scrutineers will look for any conductive components which are in close proximity (within 100mm) of any tractive system component (HV) element, which could become live in the event of a failure of a component, breach of insulation or similar either from normal operation, or in the event of a collision.

Any wire, braid or similar used for vehicle grounding must be appropriately terminated, robustly mounted, and of sufficient size and mechanical strength for its purpose.

This test involves using a Fluke 289 Multimeter on the lo-ohm setting to ensure there is a low resistance connection from the item being tested in reference to the chassis ground test point. It is acceptable to subtract the resistance of the test leads if necessary. Teams may remove paint, resin or other coverings to expose a bare conductive surface if required.

Teams also have the option of measuring resistance via method of voltage drop using a 1 amp current source if necessary.

Measurements to the conductive component should be taken within 100mm of the tractive system or as close as practical. Teams may remove body panels and covers to demonstrate compliance if required.

The pass criteria are:

a. Electrically conductive parts 300 mOhms Examples: parts made of steel, (anodized) aluminum, any other metal parts

			Criteria		Reinspect	Passed
SI.18.1	Equipotential Bonding	Firewall(s) Measure with Multimeter using GND TP and a probe	300 mOhm			
SI.18.2		Accumulator Container(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.3		Components in proximity of Accumulator Container(s) Measure with Multimeter using GND TP and a probe, this would be the TS Stack, HIP and Motor	300 mOhm 5000 mOhm			
SI.18.4		Motor Controller Enclosure(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.5		Components in proximity of Motor Controller(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.6		Motor Casings, Mounts and Scatter Shields Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.7		Components in proximity of Motor(s) Measure with Multimeter using GND TP and a probe, this includes the left rear suspension	300 mOhm 5000 mOhm			

EV CLASS - EV STATIC INSPECTION

VEHICLE VISUAL INSPECTION CONTINUED...			Criteria		Reinspect	Passed
SI.18.8	Equipotential Bonding	Other HV enclosure(s) Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.9		Components in proximity of other HV enclosures Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.10		Conductive HV cable channels, cable Armor and cable shielding Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.11		Components in proximity of HV Wires and Cables Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.12		Components in proximity of MSD Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.13		Components in proximity of the Tractive System Measurement Points Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.14		Components in proximity of the Energy Meter Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.15		Cooling system components including heatsinks, radiators, pumps and reservoirs Measure with Multimeter using GND TP and a probe	300 mOhm 5000 mOhm			
SI.18.16		Seat Mounts, Seatbelt Mounts, Driver Controls, Pedals or items in the Drivers Area if in proximity to HV components N/A	300 mOhm 5000 mOhm			
SI.18.17		Other items:	300 mOhm 5000 mOhm			
SI.18.18		Other items:	300 mOhm 5000 mOhm			
SI.18.19		Other items:	300 mOhm 5000 mOhm			
SI.18.20		Other items:	300 mOhm 5000 mOhm			
SI.18.21		Other items:	300 mOhm 5000 mOhm			

2025 Formula SAE-A Technical Inspection Sheet
EV FUNCTIONAL TEST

Initial Presentation Scrutineer Name:		**Initial Presentation Lane Number:**	
1st Reinspection Scrutineer Name:		**1st Reinspection Lane Number:**	
2nd Reinspection Scrutineer Name:		**2nd Reinspection Lane Number:**	
3rd Reinspection Scrutineer Name:		**3rd Reinspection Lane Number:**	

PERSONNEL			**Inspection #**			
Teams are required to have an Electrical Safety Officer.			**Initial ins.**	**1st reins.**	**2nd reins.**	**3rd reins.**
FT.1.1	Identify the Electrical Safety Officer (ESO)	Ask for the ESO				
FT.1.2	Only team members involved in technical inspection can be in the technical inspection bay	Ask for non-essential people to leave the area				
DO NOT PROCEED WITH TECHNICAL INSPECTION IF THE ESO IS NOT PRESENT						

INERT VEHICLE (EV & AV CLASSES ONLY)			**Inspection #**			
The Vehicle must be in a condition which prevents unexpected energization of the HV electrical systems or movement of the tractive system			**Initial ins.**	**1st reins.**	**2nd reins.**	**3rd reins.**
FT.1.3	Disable the HV Systems - Manual Service Disconnect (MSD)	Check the MSD is removed				
FT.1.4	Disable the HV Systems - Tractive System Master Switch (TSMS) Lock	Check the TSMS Lock is fitted and effective				
FT.1.5	(AV and Dual Purpose only) Disable the ASMS - Autonomous System Master Switch (ASMS) Lock	Check the ASMS Lock is fitted and effective				
DO NOT PROCEED WITH TECHNICAL INSPECTION IF MSD IS INSTALLED OR THE TSMS LOCK IS MISSING						

INSTRUCTIONS FOR SCRUTINEERS

This section applies to Electric Vehicles (EV) and Autonomous Vehicles (AV) and Dual Purpose Vehicles.
If an item is acceptable, initial the "Passed" column.
If an item is unacceptable, initial the "Reinspect" column and inform the team members they have not meet the requirements for that item.
Teams are allowed to take their vehicle away from technical inspection and address any issues before representing for reinspection.
Scrutineers may continue with the inspection when unacceptable items are found, provided it is safe to do so.
Teams are required to pass all items before the vehicle can proceed to the Brake Test or compete in Dynamic Events.

Scrutineers reserve the right to refuse to proceed with Technical Inspection if a sufficiently serious issue which might jeopardize the safety of the technical inspection team, event personal or any other persons is found at any time.

EV CLASS - FUNCTIONAL TEST

- ESO (Lukes), Jayson, Sam, Bunny (document hander)
- MSD Disconnected (In ESO's possession)
- TSMS Lock Installed (Keys in ESO's possession)
- Car in Race Ready Condition (accumulator installed, everything connected, firewall and body kit on
- Push Bar
- High Stands x 2
- Red SCA Jack Stands x 4
- Red toolbox and large cable crimps (blue handle)
- Spare Zip Tie
- Tech Box which samples, documents and tech bibles
- Fluke Multimeter with probes and banana jack proves
- Multiple hands to lift the car and remove the body kit
- Chungus Laptop with Candapter and Kvaser. CAN KING and BMS software open
- Impact Gun with Battery and 21mm Deep Impact Socket
- Test Resistor for IMD
- Test loom for BSPD

Ask if Car can be put on the small stands immediately after entering the bay and then ask if we are able to take the rear wheels off the car as it is rear wheel drive.

If they are happy with us to, ask if we can then remove the body kit and top firewall. Place the body kit onto the stands

EV CLASS - FUNCTIONAL TEST

EV FUNCTIONAL TEST PREREQUISITES This section of the Technical Inspection process is intended to ensure that the vehicle's mandated safety systems function as required by the rules. The order of testing in this section is important as the safety of some of the later tests relies on the correct functioning of systems already tested. As such there is a hold point after the testing of the safety shutdown systems. If the safety shutdown systems do not function as intended, it will be necessary to stop testing and for teams to correct any issues. Because this section requires the vehicle to be powered on, and will require the wheels to turn under power, the vehicle must have successfully passed all previous steps before proceeding. The Scrutineer must check that the vehicle has passed all required steps before proceeding. The Technical Inspectors will supply a Fluke 289 Multimeter and insulated banana test leads.				**Reinspect**	**Passed**
FT.2.1	Vehicle ready for Functional Safety Test	The vehicle must have successfully passed the following inspection stages, with all reinspect items addressed and passed. - Accumulator Inspection - EV Electrical Inspection - Mechanical Inspection - Driver Egress	Check page 2 of this document and the stickers on the vehicle's nosecone		
FT.2.2	Race Ready Condition	The vehicle must be fully assembled with the Accumulator in place and generally in race ready condition when presented to technical inspection. Present car in Race Ready Condition	Inspect		
FT.2.3	Technical Inspection Access	The vehicle must be lifted/jacked onto stands such that all of the vehicle's wheels are at least 100 mm and no more than 300mm above the ground. The driven wheels must be removed - It must be possible to freely rotate the vehicle's wheels/hubs without the vehicle moving. Stands must be robust, and the vehicle must be secure. Use red SCA car jack stands on the first height setting to support the car in the air. Use a tape measure to display its height off the ground from the wheels. Body Kit Stays on	Inspect		
FT.2.4	Line of Fire	The minimum possible number of team members and Scrutineers should be in the area when testing is being conducted. Keep people away from rotating wheels, hubs or other moving parts. Check tires for debris that could be thrown or remove wheels from vehicle. Use the impact gun with a 21mm impact deep socket to remove the wheel nuts on the rear two wheels and remove the wheels.	Inspect		
DO NOT PROCEED WITH THE FUNCTIONAL SAFETY TEST IF ANY OF THE ABOVE ARE NOT COMPLETE					
FT.2.5	Driver Egress	If any of the following tests requires a driver to be in the vehicle, the driver must be in full race gear, and it must be possible for the driver to egress the vehicle safely. e.g. The vehicle must not be on high stands. We do not need a driver in the vehicle, we will be actuating the pedals from outside the car	Inspect		

EV CLASS - FUNCTIONAL TEST

BMS FUNCTION TEST				Reinspect	Passed
FT.3.1	Accumulator Management System Function	BMS must monitor the cell voltage of each cell Display using the candapater and BMS software on the Chungus the voltage of all 108 cells in the live data tab, ask if it okay to remove the body kit to access the expansion port on the CEN and to turn GLVMS on	Activate BMS system and show measurement data of the BMS for each cell (e.g., via laptop)		
FT.3.2		BMS must monitor the temperature of at least 20% of cells in pack Using the kvaser and CAN KING on the Chungus, display the max, min and average temps of the cells in the segments. Disconnect the candapater and connect the kvaser. Display a CANaMONS PCB and explain how the board connects to the thermistors on each Energus Module, converts the analogue reading into a CAN message and sends the messages the LVD on an isolated CAN network. The LVD then sends the messages on CAN 1 which is connected to the rest of the car and the BMS		p.g.1	
FT.3.3		A red indicator light marked "BMS" must be installed in the cockpit that lights up, if the BMS shuts down the car Point out the BMS light on the MoTeC which lights up when the BMS OK High Signal goes bad	Inspect		

TRACTIVE SYSTEM ENERGISATION TEST				Reinspect	Passed
ASK THE TEAM TO INSTALL THE MSD AND UNLOCK THE TSMS LOCKOUT. THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.4.1	GLVMS Function	TS only allowed to be powered up, when GLVS is powered up. Connect the Fluke 289 Multimeter to TSMPs and set to measure DC Voltage (600V range). Try to switch on Tractive System with GLVS Master switch in Off-Position. No voltage above 60VDC allowed at measurement points. Demonstrate whilst explaining what you are doing.	Ask team to demonstrate		
FT.4.2		TS only allowed to be powered up, when GLVS is powered up. Switch on Tractive System and then switch off GLVS Master switch. Tractive system must switch off as well - Watch measured voltage on scrutineer's Multimeter fall to below 60V DC. Demonstrate whilst explaining what you are doing.	Ask team to demonstrate		
FT.4.3	Pre-Charge Circuit	A circuit that is able to pre-charge the intermediate circuit to at least 90% of the current accumulator voltage before closing the second IR has to be implemented. Use the graphing function of the Fluke 289 Multimeter to plot the voltage at the TSMPs during power up of the tractive system. Show that 90% voltage is reached before final IR closes. Demonstrate whilst explaining what you are doing. Pack will be charged to 50% soc (400ish V) and listen for the second IR to close	Ask team to demonstrate		
FT.4.4	Tractive System Voltage	The maximum tractive system voltage must be less than or equal to 600VDC at all times as measured by the Fluke 289 Multimeter. Measured Voltage: Demonstrate whilst explaining what you are doing. Should be around 400	Ask team to demonstrate		
FT.4.5	Accumulator Active Indicator	Accumulator Indicator Light or analogue voltmeter has to show if voltage above 60VDC is present outside of the accumulator container. Switch off the GLVMS. The Accumulator active indicator must continue to function even if the GLVMS is switched off or if the accumulator is removed from the vehicle. Demonstrate whilst explaining what you are doing. The AIL will remain on until the circuit has discharged below 60 VDC	Ask team to demonstrate		

EV CLASS - FUNCTIONAL TEST

TRACTIVE SYSTEM ENERGISATION TEST				Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.4.6	TSAL	The TSAL must be installed immediately under the highest point of the roll hoop. The TSAL must not be on top of the roll hoop. The TSAL must not touch the driver's helmet in any seating position. Display	Inspect		
FT.4.7		The TSAL must be continuously illuminated green in colour when the vehicle GLVMS is switched on but the voltage outside the accumulator is less than 60VDC Display, may need to use the fluke to measure TS voltage to prove it is green under 60 VDC	Inspect		
FT.4.8		Switch on the Tractive System. The TSAL must flash red in colour at between 2hz and 5Hz when the voltage outside the accumulator is more than 60VDC. The TSAL may stop functioning if the GLVMS is switched off. Display the frequency of the TSAL flash using a timer	Inspect		
FT.4.9		The TSAL must be clearly visible in bright sunlight to a person standing anywhere less than 3m from the vehicle with an eye height of 1.6m. The TSAL must not be blocked by wings or other bodywork. Minor obstruction from the roll hoop is acceptable. Display	Inspect		
FT.4.10		TSAL must be non-programmable. No part of the TSAL function must depend on a programmable device. Display TSAL schematic, prepare to demonstrate the LV/HV Separation if asked.	Ask Team to demonstrate	p.g.2	
FT.4.11	Discharge Circuit	A circuit that is able to discharge the tractive system components outside of the accumulator container to less than 60V DC withing 5 seconds of a shutdown must be fitted. Use the graphing function of the Fluke 289 Multimeter to plot the voltage at the TSMPs during power up of the tractive system. Show that the voltage falls to less than 60V DC in less than 5 seconds after the TSMS is switched off. Demonstrate whilst explaining what you are doing, have a timer to time the 5 seconds. If they ask to see a schematic, p.g.1 in EV STATIC displays the discharge circuit, or the TSAL schematic in the Electrical schematic book	Ask team to demonstrate		
FT.4.12	PDOC	A red indicator light marked "PDOC" must be installed in the cockpit that lights up, if the PDOC system shuts down the car. The indicator light must be visible to the driver in bright sunlight. The PDOC may be omitted if the pre-charge and discharge circuit is designed for continuous operation in a faulted state and will not adversely affect nearby devices Point out light on MoTeC that turns on when the PDOC OK High Signal goes bad as our PDOC is not continuous. Offer to turn the light on for them by turning the car off, disconnecting the accumulator LV loom and turning the car back on. Switch the car off to reconnect. Can show HFR schematic if asked	Inspect	p.g.3	

EV CLASS - FUNCTIONAL TEST

INSULATION MONITORING DEVICE TEST

The IMD is one of the primary methods for monitoring the health of the accumulator and tractive system.
The IMD works by continuously measuring the insulation resistance between the tractive system and the chassis. If it detects a low resistance path between the tractive system and the chassis, it will shut down the vehicle. The IMD is intended to form a detection system, providing early warning to the team that they have an insulation breakdown before a short circuit can form.
The correct function of the IMD is critical to the operational safety of the vehicle and any misbehaviour should be reported to the EV Technical Inspection Captain for investigation.

THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES

				Reinspect	Passed
FT.5.1	IMD	A red indicator light marked "IMD" must be installed in the cockpit that lights up, if the IMD system shuts down the car. The indicator light must be visible to the driver in bright sunlight. Point out light on MoTeC that turns on when the IMD OK High Signal goes bad. Offer to turn the light on for them by turning the car off, disconnecting the accumulator LV loom and turning the car back on. Switch the car off to reconnect.	Inspect		
FT.5.2	IMD	The team must provide a test resistor manufactured with insulated banana plugs with an appropriate resistor in place. The resistor must have an appropriate voltage and power rating and be encased in a hard-shell enclosure. The resistor should have appropriate power rating and be installed in a hard-shell enclosure (heat shrink/tape insulation is not acceptable). Check the resistance of the test lead with the Fluke 289 Multimeter. Display The resistance should be a minimum of: R_Test = (maximum TS voltage * 250Ω/V) - (2 * BPR) Measured Resistance: Should be (453.6V*250Ω/V) – (2*15kΩ) = 83.4kΩ *BPR = the value of the Body Protection Resistors in Ohms Installed on HIP if asked show schematic, either in electrical schematic book or pg1 of EV Static.	Inspect		
FT.5.3	IMD	Activate Tractive System and connect R_Test between HV+ and GLVS ground. TS voltage as measured at the TSMPs must decrease below 60VDC in 5 sec, IMD may take up to 30s to react. Observe with the Fluke 289 Multimeter. Demonstrate whilst explaining what you are doing, if they do not have a stopwatch, use a phone to time	Check Shutdown Time with Stopwatch		
FT.5.4	IMD	Activate Tractive System and connect R_Test between HV- and GLVS ground. TS voltage as measured at the TSMPs must decrease below 60VDC in 5 sec, IMD may take up to 30s to react. Observe with the Fluke 289 Multimeter. Demonstrate whilst explaining what you are doing, if they do not have a stopwatch, use a phone to time	Check Shutdown Time with Stopwatch		
FT.5.5	IMD	Remove the test resistor. The tractive system may not automatically return to active state after the IMD test resistor was removed or a BMS error disabled it. The Driver must not be able to reactivate the tractive system Demonstrate whilst explaining what you are doing, try to turn the TSMS switch on and off, and explain that the only way to re active the TS is to power cycle LV or press the Hard Fault Reset button on the CEN which is mounted behind the firewall. Display the HFR board schematic	Ask team to demonstrate	p.g.3	

EV CLASS - FUNCTIONAL TEST

SHUTDOWN SYSTEM TEST

This section proves the correct function of all shutdown devices.
The tractive system must fall below 60V within 5 seconds when the shutdown system is activated.
With the Fluke 289 Multimeter monitoring the voltage at the TSMPs, have the team turn on the vehicle's LV and Tractive System then activate the shutdown mechanism.
While completing this test, the items previously must continue to operate. Watch that the TSAL, Accumulator Active Indicator, precharge and discharge systems all continue to work correctly.

THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES
TAKE CARE NOT TO OVERHEAT THE PRECHARGE OR DISCHARGE RESISTORS DURING TESTS

				Reinspect	Passed
FT.6.1	Shutdown System Test	Vehicle with tractive system live --> GLVS master switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate		
FT.6.2		Vehicle with tractive system live --> TS master switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate		
FT.6.3		Vehicle with tractive system live --> Left shutdown button off Demonstrate whilst explaining what you are doing	Ask team to demonstrate		
FT.6.4		Vehicle with tractive system live --> Right shutdown button off Demonstrate whilst explaining what you are doing	Ask team to demonstrate		
FT.6.5		Vehicle with tractive system live --> Cockpit shutdown button off Demonstrate whilst explaining what you are doing	Ask team to demonstrate		
FT.6.6		Vehicle with tractive system live --> Brake over travel switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate		
FT.6.7		Vehicle with tractive system live --> GLVS master switch off Demonstrate whilst explaining what you are doing	Ask team to demonstrate		
FT.6.8		Unmount inertia switch. Activate TS and shake the switch and check if TS is shutdown. TS is not allowed to reactivate without a manual reset e.g. by the driver Demonstrate whilst explaining what you are doing, use snips to cut the zip tie to demount the inertia switch and shake it, explain that the only way to reset it is to push the switch down and is out of reach from the driver. Remount with a zip tie.	Ask team to demonstrate		

EV CLASS - FUNCTIONAL TEST

READY TO DRIVE TEST

This section tests the on-board systems which are intended to prevent the vehicle running away if a component fails.
The plausibility systems are expected to have a high degree of reliability and robustness as they may be the last line of defence from a collision.
This section will require that the team put the vehicle into drive, so the wheels and hubs will turn under power. Keep all unnecessary people outside of the test bay and ensure that people are well clear of the any rotating components.
The team may elect to have a driver inside the vehicle for this section if required. The driver needs to be in full race gear and must be able to egress the vehicle safely in the event of a fire, so pay attention to the height and stability of the vehicle and any obstructions in the inspection bay.

THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES

				Reinspect	Passed
FT.7.1	Ready to Drive	Only closing the shutdown circuit must not set the car to ready-to-drive mode. The car is ready to drive as soon as the motor(s) will respond to the input of the torque encoder/ acceleration pedal Demonstrate whilst explaining what you are doing, put the car into HV and press the accelerator pedal and explain there is no motor response as RTD is not enabled.	Ask team to demonstrate		
FT.7.2		Additional actions are required by the driver to set the car to ready-to-drive-mode e.g. pressing a dedicated start button, after the tractive system has been activated. One of these actions must include the brake pedal being pressed as ready-to-drive-mode is entered. It must be possible for the vehicle to be powered on, with the high voltage on, but with the vehicle not in drive. It is not acceptable for the action of the driver putting the vehicle in drive to simultaneously turn on the HV System and engage drive. Demonstrate whilst explaining what you are doing, explain that pressing the brake and the RTD button on the DEN will enable the car, ask that you can turn on HV and go into RTD, gentle spin the wheels.	Ask team to demonstrate		
FT.7.3		The car must make a characteristic sound, once but not continuous, for at least 1 second and a maximum of 3 seconds when it is ready to drive The sound level must be a minimum of 70dBA, fast weighting, in a radius of 2m around the car The used sound must be easily recognizable. No animal voices, song parts or sounds that can be interpreted as offensive will be accepted Demonstrate whilst explaining what you are doing. Put the car into ready to drive to display the noise. Offer to time if they do not have a stopwatch. If they wish to see a datasheet for the sounder, it can be found in the datasheet folder.	Ask team to demonstrate		
FT.7.4		Pressing the accelerator gently must cause the vehicle's wheels to rotate. Check all of the vehicle's driven wheels turn in the correct direction. Demonstrate whilst explaining what you are doing, explain we have a locked rear axle and gently spin the wheels using the accelerator (wheels off). They will spin in the right direction.	Ask team to demonstrate		

EV CLASS - FUNCTIONAL TEST					
DRIVER INPUT PLAUSIBILITY CHECKS				**Reinspect**	**Passed**
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.8.1	Torque Encoder / Brake Pedal Plausibility Check	Torque encoder is at more than 25% and brake is actuated simultaneously. The motors have to shut down. The motor power shut down has to remain active until the torque encoder signals less than 5% pedal travel, no matter whether the brake pedal is still actuated or not. 1. Ask the team to spin the motors by pressing the accelerator pedal. 2. While the wheels are spinning, ask the team to depress the brake pedal. 3. The wheels must stop turning and the torque (current) to the motors should immediately fall to zero 4. Ask the Team to release the brake pedal while continuing to hold the accelerator pedal down. The wheels must not turn. 5. Ask the team to completely release the throttle pedal and press it again. 6. The wheels may now turn. Demonstrate whilst explaining what you are doing, follow the above instructions	Ask team to demonstrate		
FT.8.2	Torque Encoder Plausibility Check	If an implausibility occurs between the values of two torque encoder sensors the power to the motor(s) has to be immediately shut down completely. It is not necessary to completely deactivate the Tractive System, the motor controller(s) shutting down the power to the motor(s) is sufficient Implausibility is defined as a deviation of more than 10% pedal travel between the sensors If three sensors are used at least two sensors have to be within 10% pedal travel, etc. 1. Check that driven axles turn by pressing the accelerator pedal 2. Disconnect at least 50% of the sensors and check that the power to the motors is shut down The sensor should be disconnected while the axles are turning. There must be no uncommanded throttle surges when the sensors are unplugged. Demonstrate whilst explaining what you are doing, follow the above instructions	Ask team to demonstrate		
FT.8.3		Repeat the above disconnecting the other throttle sensors. Demonstrate whilst explaining what you are doing, follow the above instructions	Ask team to demonstrate		

EV CLASS - FUNCTIONAL TEST					
DRIVER INPUT PLAUSIBILITY CHECKS				**Reinspect**	**Passed**
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.9.1	Brake System Plausibility Device (BSPD)	A standalone non-programmable circuit must be used on the car such that when braking hard (without locking the wheels) and when a positive current is delivered from the motor controller (a current to propel the vehicle forward), the IRs will be opened. The current limit for triggering the circuit must be set at a level where 5kW of electrical power in the DC circuit is delivered to the motors at the nominal battery voltage. The action of opening the IRs must occur if the implausibility is persistent for more than 0.5s. The team must devise a test to prove this required function during Electrical Tech Inspection. However, it is suggested that it should be possible to achieve this by sending an appropriate signal to the non-programmable circuit that represents the current to achieve 5kW whilst pressing the brake pedal to a position or with a force that represents hard braking. The preferred method for testing the BSPD system involves passing actual current from an external supply through the tractive system current sensor while the driver presses the brake pedal. Alternative test methods will require approval from the EV Technical Inspection Captain. The test must be an end-to-end test, meaning it is not acceptable for the team to simulate the 5kW by simulating the output of the current sensor, or using a substitute current sensor. To demonstrate, use the BSPD test loom and a power supply to put current into the coil plug on the side of the HIP. Display the BSPD circuit in the electrical schematic booklet and the image of the current sensor coiled and the current sensor. When at 1.25 A (10 turns on the sensor means that is 12.5A which 12.5A * 388.8 V = 4.86 kW) and the brake is pressed, the BSPD will fault, indicating on the MoTeC. This test can be done in or out of HV. EXPLAIN EVERYTHING YOU ARE DOING	Visual Inspection If required, ask team to show BSPD Schematic. Ask team to demonstrate	p.g.4	
FT.9.2		The Brake Plausibility Device may only be reset by power cycling the GLVS Master Switch or via a RESET button, located out of reach from the driver Display that the car will not go into HV now without power cycling GLVMS or pressing the HFR on the CEN which is mounted to the back of the firewall.	Ask team to demonstrate	p.g.3	

EV CLASS - FUNCTIONAL TEST

ENERGY METER TEST The energy meter test needs to be done by the Energy Meter Captain.				Reinspect	Passed
THE VEHICLE SHOULD BE ASSUMED TO BE LIVE AT ALL TIMES					
FT.10.1	Energy Meter Test	Energy Meter Captain has laptop connected Follow instructions	Ask team to demonstrate		
FT.10.2		Switch on the GLVMS and check the datalogger will communicate with the laptop Follow instructions	Ask team to demonstrate		
FT.10.3		Switch on the tractive system and rotate the wheels such that sufficient power is delivered to the motors to register on the energy meter. Follow instructions	Ask team to demonstrate		
FT.10.4		The energy meter must show both positive voltage and correct current flow direction. Follow instructions	Ask team to demonstrate		
ASK THE TEAM TO TURN OFF THE VEHICLE, REMOVE THE MSD AND LOCK THE TSMS LOCKOUT.					
SEAL TS COMPONENTS				**Reinspect**	**Passed**
FT.11.1	Seal Tractive System components	Seal all important parts with tamper proof tape after the IMD test was passed successfully: i.e. Accumulator container, motor controller housings and any other enclosures containing Tractive System Voltage. Record items sealed:	Seal Items		